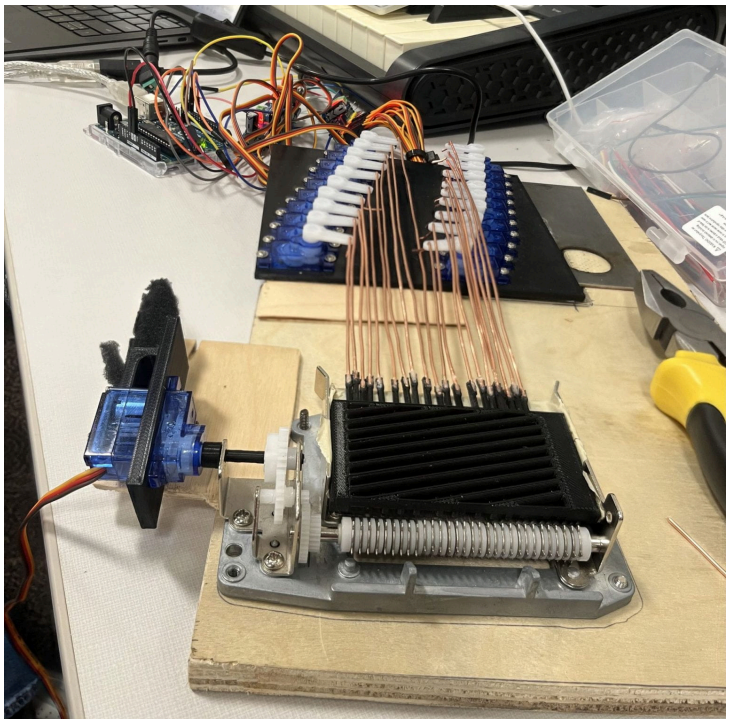
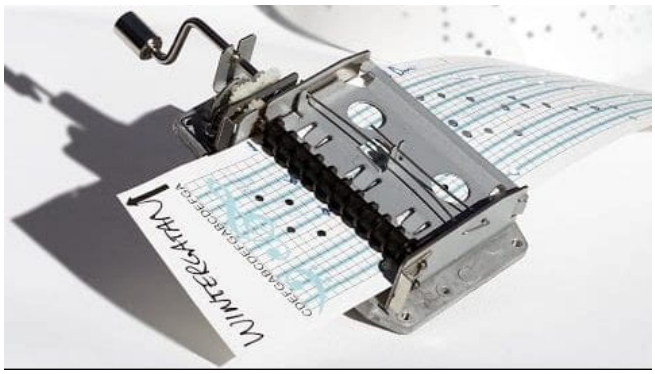
MIDI Music Box

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Making Music with Machines – A Term 2025



Our system began as a standard punch card music box mechanism. Instead of relying on paper strips with pre-punched notes, we replaced the card reader with an array of miniature SG90 servos, each linked to the box's individual tines through custom 3D-printed actuators spaced 2mm apart and coupled by brass rods. This allows each note to be struck independently, enabling full digital control.



Original Punch Card Music Box – Before Modification



After Modification

An Arduino microcontroller acts as the interface between the MIDI input and the actuators. Incoming MIDI messages are parsed and mapped to corresponding servo movements, translating note data into real mechanical plucks. Each servo moves just enough to push its 3D-printed bar against a tine, producing a note in the same way the punch card would, but now under software control. There is also a continuous servo rotating the plucker cylinder, resetting the tines after each note.

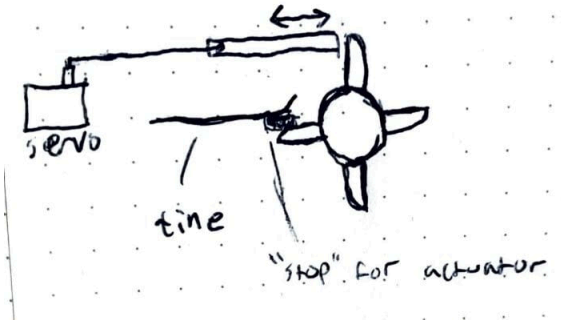


Diagram of one actuator module

The system is built around an Arduino microcontroller connected to PCA9685 servo driver boards, which expand the number of controllable servo channels. We chose to use 20 actuators instead of all 30 possible notes on the original mechanism to reduce complexity while keeping the system reliable. The active range is tuned to a C1–A3 Mixolydian scale.

Because the position of each actuator also needed to be very exact in the actuator direction, we had to tune a custom position offset for each servo. Solving this problem in software allowed us to be less precise when bending and assembling the brass rods.

```
int servoOffset[] = {-5, 15, 0, 10, 0, -5, 5, 0, 5, 20,
                     0, 5, 10, 10, 0, 10, 17, 5, 30, 0}; //offset for each individual servo
```

Custom position offset for each servo motor

In the future, we would love to experiment with solid wood board instead of plywood to get a larger sound out of the music box. Music boxes rely heavily on their resonator to produce a sound, as they are very quiet on their own. We also need a better way to isolate the loud servo noise from the system, as it masked the sound production we were aiming for.